TrustNav

Can the fleet believe its satellite position?

Explainable per-second trust scoring and response for GNSS spoofing and jamming in vehicle and drone fleets



trust score, for every vehicle, every second

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Tutor: Sofiane Hamrioui · Oral examination, 13 November 2026

Everything that moves trusts GNSS blindly

Nantes airport, April 2017

A jammer left running in a parked car delayed several flights; its owner was fined €2 000.

Lyon airport, July 2023

A truck driver's jammer, used to escape fleet tracking, disrupted signals until the ANFR traced it.

Baltic and conflict zones, summer 2024

About 1 500 flights a day spoofed; interference has been a daily event for aviation since 2022.

What exists today

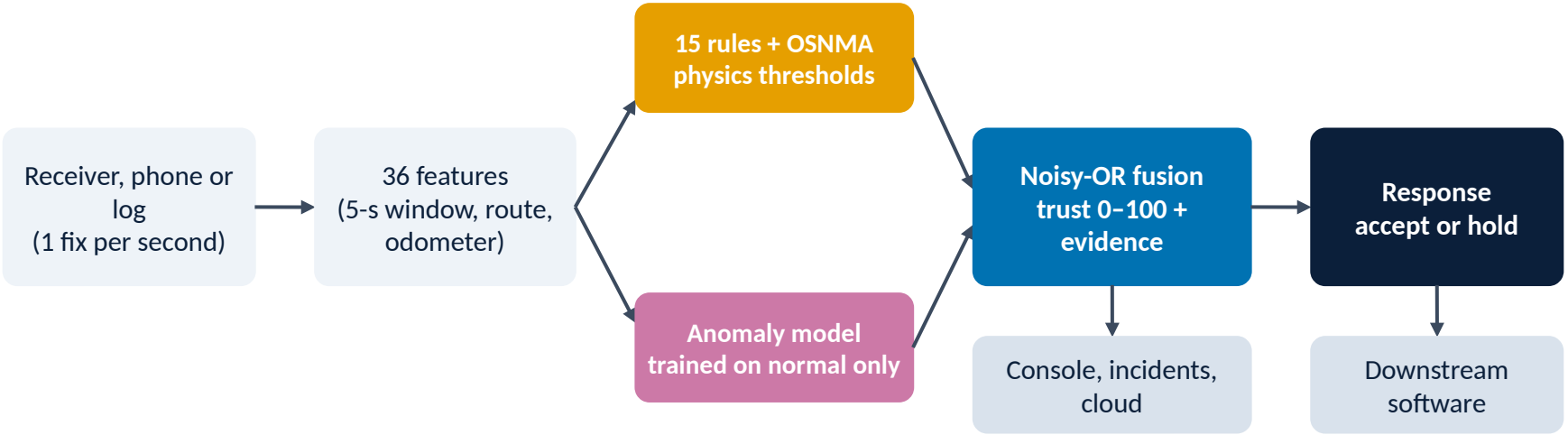
Fleet telematics — shows whatever position it is given; at best it relays the receiver's jamming flag.

Receiver flags — a state code inside one receiver, with no explanation; both flags can trip on jamming alone.

Fixed probes, anti-jam antennas — protect a site or null a jammer; they do not travel with a van or a drone.

The gap: nobody tells the operator, per vehicle and per second, whether the position is true, or acts on it.

TrustNav: a software layer beside the receiver



Scores

a trust score from 0 to 100, per vehicle and drone, every second

Explains

in plain words with values: “Fix moved 7 998 m in 1 s (limit 250 m)”

Responds

refuses the fix and tracks the vehicle by its odometer along its route

Three witnesses, one verdict

Rules

15 deterministic checks with named thresholds

- 60 m/s · 8 m/s² · a 250 m jump in one second
- 6 dB-Hz C/N0 shift · PDOP > 5 · clock drift > 60 ns/s
- critical: 4× jump or OSNMA failure → risk ≥ 70
- silent on normal data, blind to slow attacks

Anomaly model

Isolation Forest (300 trees) + per-feature tail detector

- trained on normal driving only: it never sees an attack
- calibrated tail probability p → risk = 25 · (−log10 p), capped at 100
- risk 50 = a one-in-a-hundred event under normal driving
- names the odd input: “90 m off the road, signals clean”

Galileo OSNMA

signed navigation data, operational since July 2025

- signature fails → forged data, critical
- signature passes ≠ safe: a relay attack keeps it valid
- GPS is not covered
- one witness among three, not a complete defence

Fusion by noisy-OR, not by averaging

risk = 100 · [1 − (1 − 0.6 · p rules)(1 − 0.8 · p model)] trust = 100 − risk

model 90, rules 20:
average 55 · noisy-OR 75

Response layer: the lie stops here

1. **ACCEPT** trust at least 50 and no critical rule: the fix passes through unchanged.

2. **HOLD** the fix is refused; the position handed on is dead-reckoned from the last trusted fix with the odometer, along the planned route — a spoofer can touch neither.

3. **RELEASE** after 5 clean seconds and a fix within 60 m + 3 % of the distance travelled; after 60 s with a ten-times looser bound; a fix still kilometres away stays refused.

CONTINUE	VERIFY	ISOLATE
use the fix	confirm by other means before acting	trust < 25 or critical: stop using this position



What the operator sees: score, band, the three risks and the verdict in plain words.

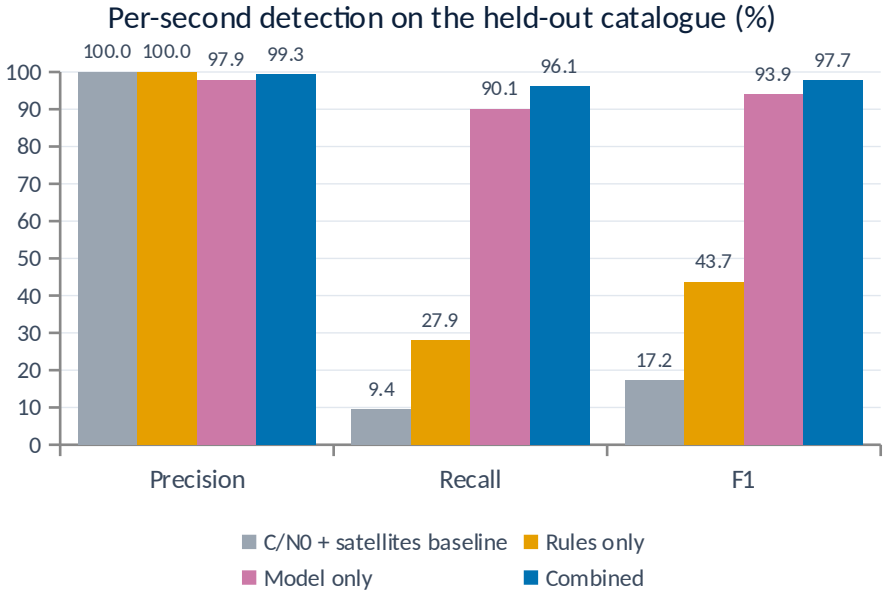


Teleport spoof on the map: the reported position (bottom right) is 8 km from the van; the protected position stays on the real path, the dashed line is the lie.

Evaluation: disjoint routes, held-out attacks

Protocol

- Real Angers street geometry; a deterministic, seeded simulator produces what a receiver would report (transmitting attacks is illegal).
- Each route has one role: 5 loops + 2 drone areas to train (normal runs only), 2 + 1 to validate (fusion weights, nothing else), 2 + 1 held out for every reported number.
- Held-out catalogue: 10 scenarios, 3 800 one-second observations, 1 550 under attack, 7 attack types.
- Alert = combined risk ≥ 50. Rule thresholds come from physics, not from data.



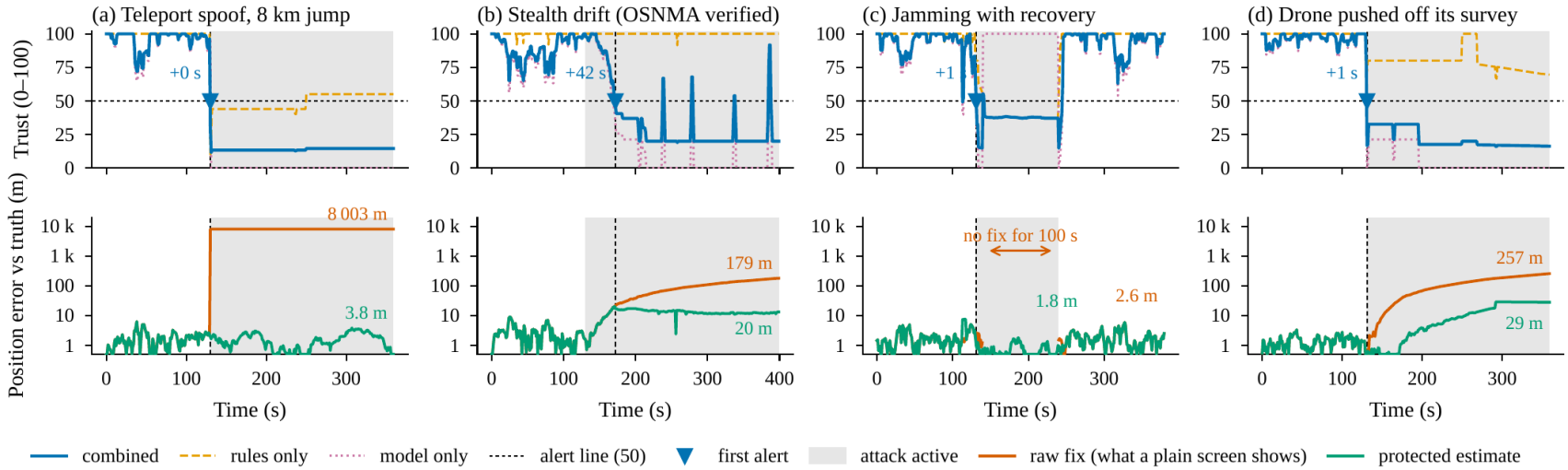
Bootstrap over scenarios, 95 % intervals

precision 97.9–100 % · recall 89.4–99.6 %
F1 0.940–0.997 · all 7 attack episodes detected
median delay 1 s · no false incident

False-alarm rate on 2 250 clean seconds:

0 % | 0 % | 1.33 % | 0.44 %

What happens, second by second



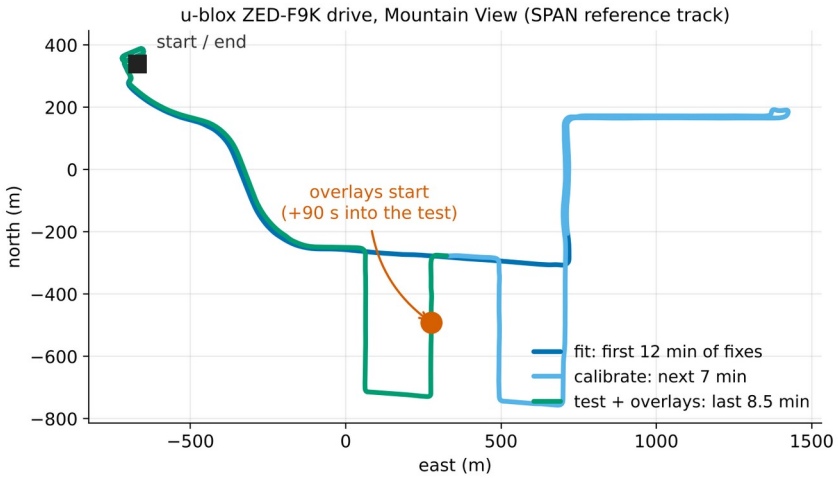
7 / 7
attack episodes detected

1 s
median delay to the first alert (42 s for the stealth drift)

29 m
worst position handed on (8.0 km on an ordinary screen)

0 s
blank screen under jamming (100 s without the response layer)

A first real receiver



Public recording: u-blox ZED-F9K with a NovAtel SPAN reference, Mountain View, February 2020 (Google gps-measurement-tools, Apache-2.0). The reference track is the route and its speed the odometer, so the protected errors are idealised.

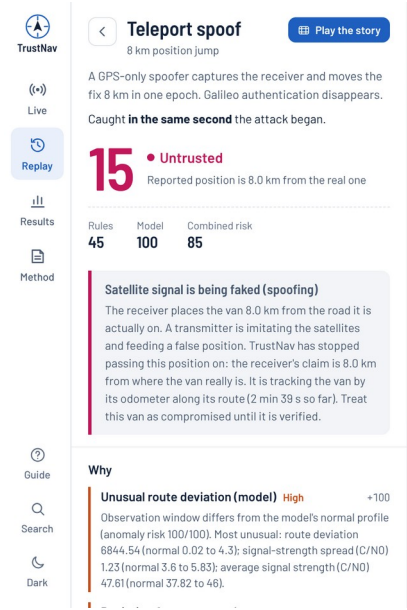
0 — **false alarms from the rules alone**
over the whole 27.6-minute drive (minimum trust 90); the C/N0 + satellites baseline raised 86 false-alarm seconds.

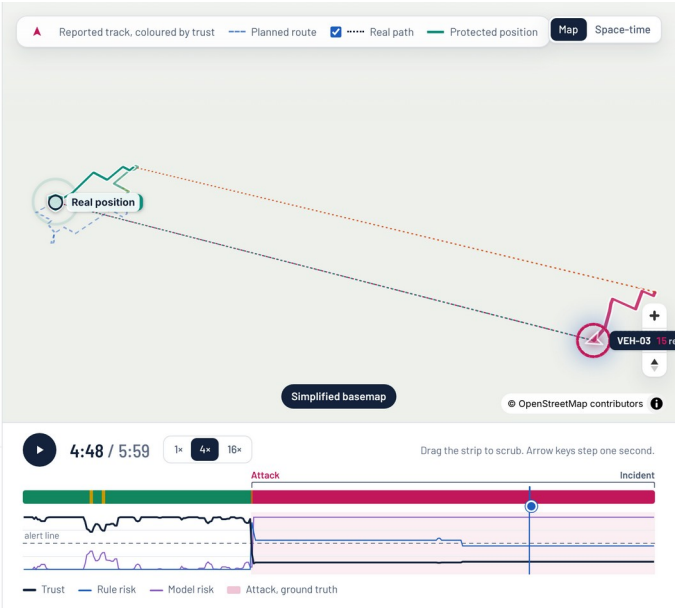
89 % — **of the seconds alerted by the simulator-trained model**
so the model must be re-calibrated on the receiver in use; re-calibrated on the first 19 minutes, it raised no alert in the last 8.5 minutes.

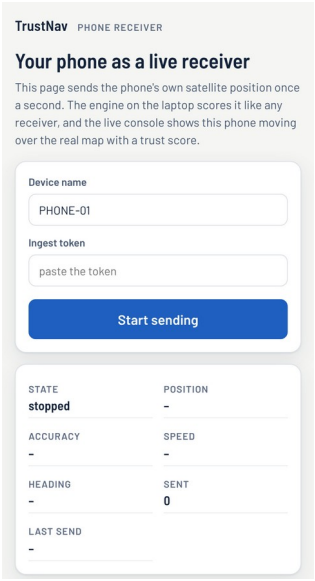
0 · 35 · 1 s — **to catch overlaid teleport, drift and jamming**
with the protected estimate within 0.8, 24 and 0.8 m of the reference.

Two weaknesses found by the real data, and fixed
tail scores saturated for extreme values (now computed in log space) · a 60-second clean streak released a teleported fix (now bounded by the rejoin tolerance)

Live demo: the console, and a phone as a receiver







Demo plan

1. Replay the teleport: score, verdict, map
2. Hold: real path vs reported lie
3. Phone: live walk, attack switch

Open source
github.com/domail2karandeshmukh-cyber/trustnav

Python engine
simulator, rules, model, response; 32 tests

Supabase cloud
row-level security, hashed ingest token, no secret in the repo

Next.js console
trustnav.vercel.app; 109 end-to-end checks on every build

What we claim, what we don't, what's next

Limits

- one simulator produces every split: the numbers are an upper bound
- route deviation assumes an assigned route
- the response is only as good as the odometer it trusts
- OSNMA simulated; no multi-antenna or IF evidence

Safety and ethics

- nothing is transmitted: every attack exists as software
- Phase 2 hardware is receive-only
- GDPR: purpose limitation, 3-day retention in the demo, transparency to drivers
- a research indicator, not certified integrity: not for aviation or maritime

Next

- hours of real drives in Angers (phone, then u-blox): false alarms per hour
- replay Jammertest 2023 and TEXBAT through a software receiver
- a second cheap receiver against synchronous spoofing
- real OSNMA from raw Galileo pages

A per-second, explained trust score and a response that keeps a lie from the systems that act on it.

Questions?

trustnav.vercel.app · github.com/domail2karandeshmukh-cyber/trustnav